

Puppet practical 1:

Setup and certificate signing

Puppet Master-Agent Setup on AWS EC2 Ubuntu 22.04 LTS (Jammy)

☒	agent	i-0ef0fc0006f197590	Running	t3.small	3/3 checks passed	ap-south-1b	ec2-3-110-41-44.ap-so...	3.110.41.44	-
☒	master	i-012bc48c9ed0f555c	Running	t3.small	3/3 checks passed	ap-south-1b	ec2-3-110-40-48.ap-so...	3.110.40.48	-

Master:

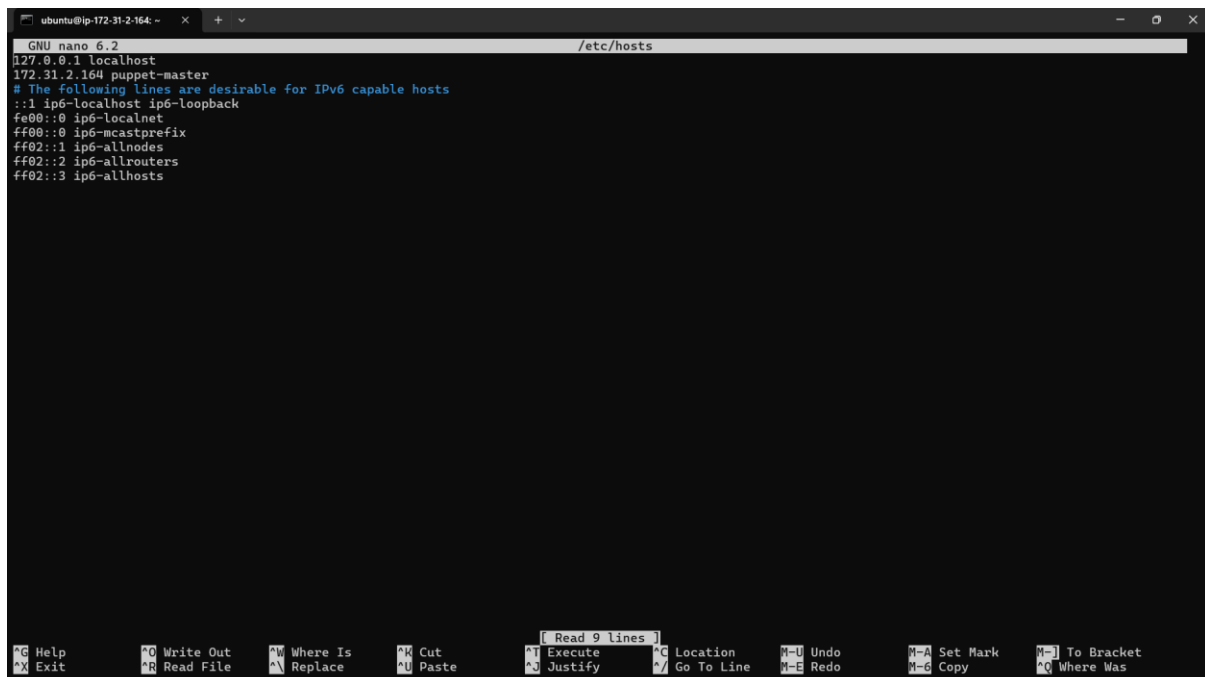
```
ubuntu@ip-172-31-2-164: ~  
Microsoft Windows [Version 10.0.26200.9278]  
(c) Microsoft Corporation. All rights reserved.  
  
C:\Users\purwa>cd Downloads  
  
C:\Users\purwa\Downloads>ssh -i "puppet-practice.pem" ubuntu@ec2-15-207-221-234.ap-south-1.compute.amazonaws.com  
ssh: connect to host ec2-15-207-221-234.ap-south-1.compute.amazonaws.com port 22: Connection timed out  
  
C:\Users\purwa\Downloads>curl ifconfig.me  
117.251.86.153  
  
C:\Users\purwa\Downloads>ssh -i "puppet-practice.pem" ubuntu@ec2-15-207-221-234.ap-south-1.compute.amazonaws.com  
The authenticity of host 'ec2-15-207-221-234.ap-south-1.compute.amazonaws.com (15.207.221.234)' can't be established.  
ED25519 key fingerprint is SHA256:gX7mJfBTKr97Yfi3tIQ812LL45Ejib0/Oy6g8t3g6FE.  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added 'ec2-15-207-221-234.ap-south-1.compute.amazonaws.com' (ED25519) to the list of known hosts.  
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1063-aws x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
System information as of Sat Sep  5 10:32:29 UTC 2026  
  
System load:  0.0          Processes:            103  
Usage of /:   23.0% of 7.57GB   Users logged in:     0  
Memory usage: 11%          IPv4 address for ens5: 172.31.2.164  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.
```

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```
ubuntu@ip-172-31-2-164:~$ sudo hostnamectl set-hostname puppet-master  
ubuntu@ip-172-31-2-164:~$ hostname  
puppet-master
```

```
ubuntu@ip-172-31-2-164:~$ cat /etc/hosts
127.0.0.1 localhost

# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
ubuntu@ip-172-31-2-164:~$ sudo nano /etc/hosts
```



```
ubuntu@ip-172-31-2-164: ~
GNU nano 6.2 /etc/hosts
127.0.0.1 localhost
172.31.2.164 puppet-master
# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts
```

```
ubuntu@ip-172-31-2-164:~$ getent hosts puppet-master
172.31.2.164 puppet-master
ubuntu@ip-172-31-2-164:~$ |
```

```
ubuntu@ip-172-31-2-164:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-2-164:~$ |
```

```

0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-2-164:~$ wget https://apt.puppetlabs.com/puppet8-release-jammy.deb
--2026-09-05 10:45:14-- https://apt.puppetlabs.com/puppet8-release-jammy.deb
Resolving apt.puppetlabs.com (apt.puppetlabs.com)... 13.225.5.127, 13.225.5.12, 13.225.5.20, ...
Connecting to apt.puppetlabs.com (apt.puppetlabs.com)[13.225.5.127]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3786 (3.7K) [application/vnd.debian.binary-package]
Saving to: 'puppet8-release-jammy.deb'

puppet8-release-jammy.deb      100%[=====] 3.70K  --.-KB/s  in 0s

2026-09-05 10:45:14 (1.64 GB/s) - 'puppet8-release-jammy.deb' saved [3786/3786]

ubuntu@ip-172-31-2-164:~$ sudo dpkg -i puppet8-release-jammy.deb
Selecting previously unselected package puppet8-release.
(Reading database ... 66021 files and directories currently installed.)
Preparing to unpack puppet8-release-jammy.deb ...
Unpacking puppet8-release (1.0.0-10jammy) ...
Setting up puppet8-release (1.0.0-10jammy) ...
ubuntu@ip-172-31-2-164:~$ |

```

```

ubuntu@ip-172-31-2-164:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Get:4 http://apt.puppet.com jammy InRelease [113 kB]
Get:5 http://apt.puppet.com jammy/puppet8 all Packages [9208 B]
Hit:6 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:7 http://apt.puppet.com jammy/puppet8 amd64 Packages [14.8 kB]
Fetched 137 kB in 2s (87.3 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ubuntu@ip-172-31-2-164:~$ |

```

```

ubuntu@ip-172-31-2-164:~$ sudo apt install -y puppetserver
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config fonts-dejavu-core java-common libasound2 libasound2-data libavahi-client3
  libavahi-common-data libavahi-common3 libcups2 libfontconfig1 libgraphite2-3 libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2 libpcsclite1 net-tools
  openjdk-17-jre-headless puppet-agent
Suggested packages:
  default-jre libasound2-plugins alsa-utils cups-common liblcms2-utils pcsd libnss-mdns fonts-dejavu-extra fonts-ipafont-gothic fonts-ipafont-mincho
  fonts-wqy-microhei | fonts-wqy-zenhei fonts-indic
The following NEW packages will be installed:
  alsa-topology-conf alsa-ucm-conf ca-certificates-java fontconfig-config fonts-dejavu-core java-common libasound2 libasound2-data libavahi-client3
  libavahi-common-data libavahi-common3 libcups2 libfontconfig1 libgraphite2-3 libharfbuzz0b libjpeg-turbo8 libjpeg8 liblcms2-2 libpcsclite1 net-tools
  openjdk-17-jre-headless puppet-agent puppetserver
0 upgraded, 22 newly installed, 0 to remove and 0 not upgraded.
Need to get 164 MB of archives.
After this operation, 429 MB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 alsa-topology-conf all 1.2.5.1-2 [15.5 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasound2-data all 1.2.6.1-1ubuntu1.2 [19.3 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libasound2 amd64 1.2.6.1-1ubuntu1.2 [391 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 alsa-ucm-conf all 1.2.6.3-1ubuntu1.12 [43.5 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 java-common all 0.72build2 [6782 B]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libavahi-common-data amd64 0.8-5ubuntu5.5 [24.0 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libavahi-common3 amd64 0.8-5ubuntu5.5 [23.9 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libavahi-client3 amd64 0.8-5ubuntu5.5 [20.1 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libcups2 amd64 2.4.10p1-1ubuntu4.21 [266 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 fonts-dejavu-core all 2.37-2build1 [1041 kB]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 fontconfig-config all 2.13.1-4.2ubuntu5 [29.1 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libfontconfig1 amd64 2.13.1-4.2ubuntu5 [131 kB]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 liblcms2-2 amd64 2.12-rc1-2ubuntu0.1 [159 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libjpeg-turbo8 amd64 2.1.2-0ubuntu1 [134 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/main amd64 libjpeg8 amd64 8c-2ubuntu10 [2264 B]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libgraphite2-3 amd64 1.3.14-1ubuntu0.1 [71.6 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libharfbuzz0b amd64 2.7.4-1ubuntu3.2 [353 kB]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libpcsclite1 amd64 1.9.5-3ubuntu1 [19.8 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 openjdk-17-jre-headless amd64 17.0.20+8-1~22.04 [48.4 MB]
Get:20 http://apt.puppet.com jammy/puppet8 amd64 puppet-agent amd64 8.10.0-1jammy [36.0 MB]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 ca-certificates-java all 20190909ubuntu1.3 [12.3 kB]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 net-tools amd64 1.60+git20181103.0eebece-1ubuntu5.4 [204 kB]
Get:23 http://apt.puppet.com jammy/puppet8 amd64 puppetserver all 8.7.0-1jammy [76.3 MB]
82% [23 puppetserver 41.1 MB/76.3 MB 54%]

```

No VM guests are running outdated hypervisor (qemu) binaries on this host.

```

ubuntu@ip-172-31-2-164:~$ sudo nano /etc/default/puppetserver
ubuntu@ip-172-31-2-164:~$ |

```

```
ubuntu@ip-172-31-2-164: ~$ nano /etc/default/puppetserver *
GNU nano 6.2 /etc/default/puppetserver *
#####
# Init settings for puppetserver
#####
# Location of your Java binary (version 8)
JAVA_BIN="/usr/bin/java"
# Modify this if you'd like to change the memory allocation, enable JMX, etc
JAVA_ARGS="-Xms1g -Xmx1g -Djruby.logger.class=com.puppetlabs.jruby_utils.jruby.Slf4jLogger"
# Modify this as you would JAVA_ARGS but for non-service related subcommands
JAVA_ARGS_CLI="${JAVA_ARGS_CLI:-}"
# Modify this if you'd like TrapperKeeper specific arguments
TK_ARGS=""
# These normally shouldn't need to be edited if using OS packages
USER="puppet"
GROUP="puppet"
INSTALL_DIR="/opt/puppetlabs/server/apps/puppetserver"
CONFIG="/etc/puppetlabs/puppetserver/conf.d"
# Bootstrap path
BOOTSTRAP_CONFIG="/etc/puppetlabs/puppetserver/services.d/,/opt/puppetlabs/server/apps/puppetserver/config/services.d/"
# SERVICE_STOP_RETRIES can be set here to alter the default stop timeout in
# seconds. For systemd, the shorter of this setting or 'TimeoutStopSec' in
# the systemd.service definition will effectively be the timeout which is used.
SERVICE_STOP_RETRIES=60
# START_TIMEOUT can be set here to alter the default startup timeout in
# seconds. For systemd, the shorter of this setting or 'TimeoutStartSec'
# in the service's systemd.service configuration file will effectively be the
# timeout which is used.
START_TIMEOUT=300
# Maximum number of seconds that can expire for a service reload attempt before
# the result of the attempt is interpreted as a failure.
RELOAD_TIMEOUT=120
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^I Execute    ^C Location   ^U Undo       ^M Set Mark   ^_] To Bracket
^X Exit      ^R Read File  ^N Replace    ^P Paste      ^J Justify    ^_ Go To Line  ^B Redo       ^G Copy       ^_ Where Was
```

```
ubuntu@ip-172-31-2-164:~$ sudo systemctl start puppetserver
ubuntu@ip-172-31-2-164:~$ sudo systemctl status puppetserver
● puppetserver.service - puppetserver Service
   Loaded: loaded (/lib/systemd/system/puppetserver.service; disabled; vendor preset: enabled)
   Active: active (running) since Sat 2026-09-05 10:51:44 UTC; 15s ago
     Process: 4619 ExecStart=/opt/puppetlabs/server/apps/puppetserver/bin/puppetserver start (code=exited, status=0/SUCCESS)
    Main PID: 4680 (java)
       Tasks: 62 (limit: 4915)
      Memory: 1.1G
         CPU: 1min 2.448s
    CGroup: /system.slice/puppetserver.service
            └─4680 /usr/bin/java --add-opens java.base/sun.nio.ch=ALL-UNNAMED --add-opens java.base/java.io=ALL-UNNAMED -Xms1g -Xmx1g -Djruby.logger.class=
              4811 /opt/puppetlabs/puppet/bin/ruby /opt/puppetlabs/server/data/puppetserver/dropsnde/bin/dropsnde submit

Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,567 |-INFO in ch.qos.logback.core.rolling.SizeAndTimeBasedFNATP@7bdb30ef - The date pattern is 'S
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,567 |-INFO in ch.qos.logback.core.rolling.SizeAndTimeBasedFNATP@7bdb30ef - Roll-over at midnight.
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,569 |-INFO in ch.qos.logback.core.rolling.SizeAndTimeBasedFNATP@7bdb30ef - Setting initial perioS
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,570 |-INFO in ch.qos.logback.core.model.processor.ImplicitModelHandler - Assuming default type [p
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,576 |-INFO in ch.qos.logback.core.rolling.RollingFileAppender[FILE] - Active log file name: /var>
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,576 |-INFO in ch.qos.logback.core.rolling.RollingFileAppender[FILE] - File property is set to [/>
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,576 |-INFO in ch.qos.logback.core.model.processor.AppenderRefModelHandler - Attaching appender n>
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,576 |-INFO in ch.qos.logback.core.model.processor.DefaultProcessor@64cea43c - End of configurati>
Sep 05 10:51:43 puppet-master puppetserver[4680]: 10:51:43,576 |-INFO in ch.qos.logback.access.joran.JoranConfigurator@4b1c82db - Registering current confi>
Sep 05 10:51:44 puppet-master systemd[1]: Started puppetserver Service.
lines 1-22/22 (END)
```

Agent:

```
ubuntu@ip-172-31-4-252: ~  
* Documentation: https://help.ubuntu.com  
* Management:   https://landscape.canonical.com  
* Support:      https://ubuntu.com/pro  
  
System information as of Sat Sep  5 10:59:17 UTC 2026  
  
System load:  0.03      Processes:      113  
Usage of /:   23.0% of 7.57GB    Users logged in:  0  
Memory usage: 12%      IPv4 address for ens5: 172.31.4.252  
Swap usage:   0%  
  
Expanded Security Maintenance for Applications is not enabled.  
  
0 updates can be applied immediately.  
  
Enable ESM Apps to receive additional future security updates.  
See https://ubuntu.com/esm or run: sudo pro status  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
To run a command as administrator (user "root"), use "sudo <command>".  
See "man sudo_root" for details.  
  
ubuntu@ip-172-31-4-252:~$
```

```
See "man sudo_root" for details.  
  
ubuntu@ip-172-31-4-252:~$ hostname  
ip-172-31-4-252  
ubuntu@ip-172-31-4-252:~$ sudo hostnamectl set-hostname puppet-agent  
ubuntu@ip-172-31-4-252:~$ sudo hostnamectl set-hostname puppet-agent  
ubuntu@ip-172-31-4-252:~$ hostname  
puppet-agent  
ubuntu@ip-172-31-4-252:~$ |
```

```
puppet agent
ubuntu@ip-172-31-4-252:~$ sudo /etc/hosts
sudo: /etc/hosts: command not found
ubuntu@ip-172-31-4-252:~$ sudo nano /etc/hosts
```

```
ubuntu@ip-172-31-4-252: ~
GNU nano 6.2 /etc/hosts *
127.0.0.1 localhost
172.31.2.164 puppet-master
# The following lines are desirable for IPv6 capable hosts
::1 ip6-localhost ip6-loopback
fe00::0 ip6-localnet
ff00::0 ip6-mcastprefix
ff02::1 ip6-allnodes
ff02::2 ip6-allrouters
ff02::3 ip6-allhosts

Help      Write Out  Where Is  Cut       Execute   Location  Undo      Set Mark  To Bracket
Exit      Read File  Replace   Paste     Justify   Go To Line Redo      Copy      Where Was
```

```
ubuntu@ip-172-31-4-252:~$ getent hosts puppet-master
172.31.2.164 puppet-master
ubuntu@ip-172-31-4-252:~$ |
```

```
ubuntu@ip-172-31-4-252:~$ nc -vz puppet-master 8140
Connection to puppet-master (172.31.2.164) 8140 port [tcp/puppet] succeeded!
ubuntu@ip-172-31-4-252:~$ |
```

```
ubuntu@ip-172-31-4-252:~$ sudo apt update && sudo apt upgrade -y
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://security.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [3795 kB]
Get:12 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1282 kB]
Get:13 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [323 kB]
Get:14 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [30.8 kB]
Get:15 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [76.5 kB]
Get:16 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [15.9 kB]
Get:17 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [756 B]
Get:18 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [70.2 kB]
Get:19 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.4 kB]
Get:20 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [412 B]
Get:21 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]
Get:22 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.8 kB]
Get:23 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.9 kB]
Get:24 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [676 B]
Get:25 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:26 http://security.ubuntu.com/ubuntu jammy-security/main amd64 Packages [3523 kB]
Get:27 http://security.ubuntu.com/ubuntu jammy-security/main Translation-en [495 kB]
Get:28 http://security.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [6299 kB]
Get:29 http://security.ubuntu.com/ubuntu jammy-security/restricted Translation-en [1212 kB]
Get:30 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 Packages [1048 kB]
Get:31 http://security.ubuntu.com/ubuntu jammy-security/universe Translation-en [235 kB]
Get:32 http://security.ubuntu.com/ubuntu jammy-security/universe amd64 c-n-f Metadata [23.2 kB]
Get:33 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 Packages [69.1 kB]
Get:34 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [12.9 kB]
Get:35 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [544 B]
Fetched 39.3 MB in 6s (6716 kB/s)
```

```
ubuntu@ip-172-31-4-252: ~$ sudo apt update
Get:34 http://security.ubuntu.com/ubuntu jammy-security/multiverse Translation-en [12.9 kB]
Get:35 http://security.ubuntu.com/ubuntu jammy-security/multiverse amd64 c-n-f Metadata [544 B]
Fetched 39.3 MB in 6s (6716 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
ubuntu@ip-172-31-4-252:~$ wget https://apt.puppetlabs.com/puppet8-release-jammy.deb
--2026-09-05 11:07:20-- https://apt.puppetlabs.com/puppet8-release-jammy.deb
Resolving apt.puppetlabs.com (apt.puppetlabs.com)... 13.225.5.12, 13.225.5.15, 13.225.5.20, ...
Connecting to apt.puppetlabs.com (apt.puppetlabs.com)[13.225.5.12]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3786 (3.7K) [application/vnd.debian.binary-package]
Saving to: 'puppet8-release-jammy.deb'

puppet8-release-jammy.deb
100%[=====] 3.79K --.-KB/s in 0s

2026-09-05 11:07:24 (158 MB/s) - 'puppet8-release-jammy.deb' saved [3786/3786]

ubuntu@ip-172-31-4-252:~$ sudo dpkg -i puppet8-release-jammy.deb
Selecting previously unselected package puppet8-release.
(Reading database ... 66021 files and directories currently installed.)
Preparing to unpack puppet8-release-jammy.deb ...
Unpacking puppet8-release (1.0.0-10jammy) ...
Setting up puppet8-release (1.0.0-10jammy) ...
ubuntu@ip-172-31-4-252:~$ sudo apt update
Hit:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy InRelease
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:5 http://apt.puppet.com jammy InRelease [113 kB]
Get:6 http://apt.puppet.com jammy/puppet8 amd64 Packages [14.8 kB]
Get:7 http://apt.puppet.com jammy/puppet8 all Packages [9208 B]
Fetched 137 kB in 2s (84.8 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ubuntu@ip-172-31-4-252:~$
```

```
ubuntu@ip-172-31-4-252:~$ sudo apt install -y puppet-agent
Hit:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-updates InRelease
Hit:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu jammy-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu jammy-security InRelease
Get:5 http://apt.puppet.com jammy InRelease [113 kB]
Get:6 http://apt.puppet.com jammy/puppet8 amd64 Packages [14.8 kB]
Get:7 http://apt.puppet.com jammy/puppet8 all Packages [9208 B]
Fetched 137 kB in 2s (84.8 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
ubuntu@ip-172-31-4-252:~$ sudo apt install -y puppet-agent
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following NEW packages will be installed:
  puppet-agent
0 upgraded, 1 newly installed, 0 to remove and 0 not upgraded.
Need to get 36.0 MB of archives.
After this operation, 125 MB of additional disk space will be used.
Get:1 http://apt.puppet.com jammy/puppet8 amd64 puppet-agent amd64 8.10.0-1jammy [36.0 MB]
Fetched 36.0 MB in 0s (83.6 MB/s)
Selecting previously unselected package puppet-agent.
(Reading database ... 66026 files and directories currently installed.)
Preparing to unpack .../puppet-agent.8.10.0-1jammy_amd64.deb ...
Unpacking puppet-agent (8.10.0-1jammy) ...
Setting up puppet-agent (8.10.0-1jammy) ...
Created symlink /etc/systemd/system/multi-user.target.wants/pxp-agent.service → /lib/systemd/system/pxp-agent.service.
Created symlink /etc/systemd/system/multi-user.target.wants/puppet.service → /lib/systemd/system/puppet.service.
Removed /etc/systemd/system/multi-user.target.wants/pxp-agent.service.
Processing triggers for libc-bin (2.35-0ubuntu3.14) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-4-252:~$
```

```
ubuntu@ip-172-31-4-252:~$ sudo /opt/puppetlabs/bin/puppet config set server puppet-master.ap-south-1.compute.internal --section main
ubuntu@ip-172-31-4-252:~$ sudo /opt/puppetlabs/bin/puppet config print server
puppet-master.ap-south-1.compute.internal
ubuntu@ip-172-31-4-252:~$ sudo nano /etc/hosts
ubuntu@ip-172-31-4-252:~$
```

```
ubuntu@ip-172-31-4-252: ~  
GNU nano 6.2 /etc/hosts  
127.0.0.1 localhost  
172.31.2.164 puppet-master.ap-south-1.compute.internal puppet-master  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
ff02::3 ip6-allhosts
```

Master:

```
ubuntu@ip-172-31-2-164:~$ sudo /opt/puppetlabs/bin/puppetserver ca list  
Fatal error when running action 'list'  
Error: Failed connecting to https://puppet-master.ap-south-1.compute.internal:8140/puppet-ca/v1/certificate_statuses/any_key?state=requested  
Root cause: Failed to open TCP connection to puppet-master.ap-south-1.compute.internal:8140 (getaddrinfo: Name or service not known)  
ubuntu@ip-172-31-2-164:~$ sudo /opt/puppetlabs/bin/puppet config set server puppet-master.ap-south-1.compute.internal --section main  
ubuntu@ip-172-31-2-164:~$ sudo /opt/puppetlabs/bin/puppet config print server  
puppet-master.ap-south-1.compute.internal  
ubuntu@ip-172-31-2-164:~$ getent hosts puppet-master.ap-south-1.compute.internal  
ubuntu@ip-172-31-2-164:~$ sudo nano /etc/hosts  
ubuntu@ip-172-31-2-164:~$
```

```
ubuntu@ip-172-31-2-164:~  
GNU nano 6.2 /etc/hosts  
127.0.0.1 localhost  
172.31.2.164 puppet-master.ap-south-1.compute.internal puppet-master puppet  
# The following lines are desirable for IPv6 capable hosts  
::1 ip6-localhost ip6-loopback  
fe00::0 ip6-localnet  
ff00::0 ip6-mcastprefix  
ff02::1 ip6-allnodes  
ff02::2 ip6-allrouters  
ff02::3 ip6-allhosts  
[ Wrote 9 lines ]  
^G Help      ^O Write Out  ^W Where Is   ^K Cut        ^T Execute    ^C Location   ^U Undo       ^A Set Mark   ^] To Bracket  
^X Exit      ^R Read File  ^N Replace    ^U Paste      ^J Justify    ^_ Go To Line  ^E Redo       ^G Copy       ^_ Where Was
```

```
ubuntu@ip-172-31-2-164:~$ sudo /opt/puppetlabs/bin/puppetserver ca list  
Requested Certificates:  
  puppet-agent (SHA256) F5:A1:0E:9E:B2:16:75:3C:08:70:A4:DC:4A:C7:EB:75:B1:90:F7:9C:FC:59:E5:E2:C6:D1:C5:28:11:54:4B:63  
ubuntu@ip-172-31-2-164:~$ sudo /opt/puppetlabs/bin/puppetserver ca sign --certname puppet-agent  
Successfully signed certificate request for puppet-agent  
ubuntu@ip-172-31-2-164:~$
```

Agent:

```
ubuntu@ip-172-31-4-252: ~  
ubuntu@ip-172-31-4-252:~$ sudo /opt/puppetlabs/bin/puppet agent -t  
Info: Using environment 'production'  
Info: Retrieving pluginfacts  
Info: Retrieving plugin  
Notice: Requesting catalog from puppet-master.ap-south-1.compute.internal:8140 (172.31.2.164)  
Notice: Catalog compiled by puppet-master.ap-south-1.compute.internal  
Info: Caching catalog for puppet-agent  
Info: Applying configuration version '1788607871'  
Notice: Applied catalog in 0.01 seconds  
ubuntu@ip-172-31-4-252:~$
```